Research Team “RT

#Many Daughters pre-analysis plan (PAP)

Preamble to the analysis (RT

Following established research (e.g., Washington, 2008), we treat the gender of a firstborn child as a quasi-random variable[1]. While parenthood itself is a non-random and likely selective process—shaped by factors such as socioeconomic status, relationship stability, or personal values—the gender of the first child is not. Conditional on entering parenthood, the probability of having a son or a daughter is effectively random and uncorrelated with pre-existing parental characteristics. Based on this assumption, we treat parents whose first child is a daughter as statistically comparable to those whose first child is a son. Our analysis focuses exclusively on first-time parents, since the gender of the first child may influence subsequent fertility decisions (e.g., by gender-based stopping rules for fertility decisions), potentially introducing bias if later births were included. As such, our causal claims pertain only to first-time parents and, thus, should not be generalized to the broader population, including for example persons with multiple births, or childless individuals.

Given the above outlined assumptions on randomness of the gender of the first-born child, we can estimate the causal effect of having a daughter on the four outcomes of interest using simple linear regression (OLS)[2]. To account for potential dependence between observations from the same household (as we sometimes observe both parents), we use robust standard errors clustered at the household level. To further strengthen our causal identification strategy, we restrict the sample to individuals surveyed within a narrow window following the birth of their first child—typically between 1 and 12 months, or, when necessary due to limited sample size, up to 24 months. This timing restriction aims to minimize confounding by external events or evolving family dynamics that may differ systematically between parents of sons and daughters over time (and may be unobservable). While this is not a perfect solution,

[1] There is for example no evidence to suggest the existence of sex-selective abortion practices in Germany. Medical professionals are legally prohibited from disclosing the sex of a fetus before the 14th week of pregnancy. While abortions are without legal penalty up to the 14th week, terminations beyond that point are subject to criminal penalty, except in specific medical or legal circumstances. These legal and institutional conditions support the assumption that the gender of the first child can be treated as effectively random in the German context.

[2] Please note that Difference-in-Differences (DiD) models are, in principle, a viable approach given our identification assumptions. However, due to substantial time lags between measurement waves for some outcomes—and the fact that certain outcomes (e.g., time spent on care work) are only observed after the birth of the first child—a DiD design is not feasible in this context. We therefore rely on simple linear regression models throughout the analysis.

it helps reduce the risk of bias and improves internal validity. We acknowledge that this restriction comes with trade-offs: it reduces the sample size and, correspondingly, lowers statistical power. What's more is that we cannot identify possible long-term trends with the chosen strategy. Of note, the chosen window (1-12 or 24 months) is not based on a specific precedent in the literature, but rather reflects a pragmatic compromise—tight enough (and hopefully not to wide) to support a causal claim, but wide enough to retain sufficient statistical power for meaningful estimation.

To account for minor imbalances between treatment and control group and to take the stratified nature of the SOEP sample into account we adjust for a basic set of controls in all our models.

- Controls:
    - Age (in years):
        - constructed with variables gebjahr, syear; data set: ppathl
            - gen age = syear - gebjahr if gebjahr<.
    - Education (Education in categories - CASMIN):
        - based on pgcasmin; data set: pgen
            - recode pgcasmin (1/3 = 1 "Low education") (4/7 = 2 "Intermediate education") (8/9 = 3 "High Education") (0 . = 4 "other"), gen(edu)
    - Marital Status (married [=1] / other [=0]):
        - based on pgfamstd; data set: pgen
            - recode pgfamstd (1 2 6 = 1 "Married") (3 4 5 7 8 = 0 "Other"), gen(fam)
    - Gender (female [=1] / male [=0]); if applicable (males and females included in model):
        - based on sex; data set: ppathl
            - recode sex (1 = 0 "male")(2 = 1 "female"), gen(gender)
    - Region (east (= 0) / west (=1) Germany):
        - based on sampreg; data set: ppathl
            - recode sampreg (2 = 0), gen(west)
    - Employment status (employed [=1] / not employed [= 0]):
        - based on pgemplst; data set: pgen
            - recode pgemplst (1 2 3 4 6 = 1 "work") (5 = 0 "not employed"), gen(work)
    - Yearly fixed effects; if applicable (more than one wave included):
        - based on syear; data set: ppathl

Please note that, due to the comparatively high number of missing values for the variable on education, these missing have been coded so that they are part of the "other" category in the education variable.

As per the request of the project team, we provide exactly one causal estimate per hypothesis. If this were a stand-alone research paper aimed at establishing causal effects, the analysis would have been substantially more comprehensive. Specifically, we would have included formal balance tests between treatment and control groups, explored alternative thresholds for the time window between birth and interview, and conducted a range of robustness checks—including for example alternative sample specifications, and other sensitivity analyses. This approach would be guided by the understanding that causal inference is often not a binary concept, where an effect is either definitively causal or not. Rather, often causality exists on a continuum—where thoughtful design, robustness checks, and transparent assumptions help researchers move closer to identifying a plausible causal relationship. In this framework, robustness checks are essential for strengthening the credibility of findings and reinforcing the validity of causal claims. However, given the constraints of the project, presenting multiple estimates and robustness analyses is not feasible. We have therefore opted for a strict test of causality, focusing exclusively on first-time births and a narrow time window between birth and the survey interview. Despite these limitations, we believe our approach remains parsimonious yet credible, adhering to the core principles of causal inference while meeting the requirements of the project.

Plan for hypothesis testing

Hypothesis 1

Having daughters leads to stronger support for gender equality (as measured by attitudes).
That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

1. **Variables:**
    a. Outcome: Gender equality measured by support for the statement “It’s best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.” (var: plh0308_v2, data: pl). Answer scale is a 7-point likert scale ranging from 1 completely disagree to 7 completely agree.
    b. Central IV: Having as first child a daughter (1/0) constructed by using biobirth data on first-born sex (kidsex01; biobirth). The dummy takes the value of 1 if first kid’s sex is identified as female and 0 as male. Due to limited case numbers, we extend the sample to include all parents whose first child was born between 1 and 24 months prior to the 2018 interview—rather than restricting the window to 1–12 months, as done in the other hypothesis tests. We account for this longer time frame by including a control for months passed since first-born birth (msb). However, this is not without limitations: if the treatment effect evolves

over time, part of it may be captured by the msb variable, potentially attenuating the estimated effect of having a daughter.

c. Controls: see preamble

2. **Inclusion/exclusion criteria:**
   Observations: The analysis sample consists of individuals from private households (inlist(pop,1,2,5,6)) who conducted an individual questionnaire (inrange(netto,10,19)) in 2018 (syear==2018) whose first-born child was born 1 up to 24 months before the interview and was not part of a multiple birth. We assume missingness on covariates are completely at random with regard to the association of outcome and gender of first-born child thus excluding observations with missing values on either of the covariates (unless specifically coded as values, see preamble), auxiliary information used for variable construction (interview month, birth year and birth month, etc.) or dependent variable (i.e., list-wise deletion).

3. **Exact analysis:**
   a. Method: OLS regression
   b. Standard error adjustment: clustered robust standard errors at the household level (hid).

4. **Interpretation of results:**
   We are interested in both the direction and statistical significance of the "daughter" effect. To preliminarily confirm the hypothesis, the estimated coefficient must be positive and statistically significant at the 5% level or below.

Hypothesis 2

Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

1. **Variables:**
   a. Outcome: Political attitudes measured by support for the statement "In politics, people often talk about "left" and "right" when describing different political views. When you think about your own political views, how would you rate them on the scale below?" (var: plh0004, data: pl). Answer scale is a 11-point scale ranging from 0 "far left" to 10 "far right". The variable was reverse coded so that 10 represents respondents on the far left and 0 those on the far right.
   b. Central IV: Having as first child a daughter (1/0) constructed by using biobirth data on first-born sex (kidsex01; biobirth). The dummy takes the value of 1 if first kid's sex is identified as female and 0 as male.
   c. Controls: see preamble.

2. **Inclusion/exclusion criteria:**
   Observations: The analysis sample consists of individuals from private households (inlist(pop,1,2,5,6)) who conducted an individual questionnaire (inrange(netto,10,19)) in 2005, 2009, 2014 or 2019 (syear==2005 | syear==2009 | syear==2014 | syear ==2019) whose first-born child was born 1 up to 12 months before the interview and was not part of a multiple birth. We assume missingness on covariates to be completely at random with regard to the association of outcome and gender of first-born child thus excluding observations with missing values on either of the covariates (unless specifically coded as values, see preamble), auxiliary information used for variable construction (interview month, birth year and birth month, etc.) or dependent variable (i.e., list-wise deletion).

3. **Exact analysis:**
   a. Method: OLS regression
   b. Standard error adjustment: clustered robust standard errors at the household level (hid).

4. **Interpretation of results:**
   We are interested in both the direction and statistical significance of the "daughter" effect. To preliminarily confirm the hypothesis, the estimated coefficient must be positive and statistically significant at the 5% level or below.

<u>Hypothesis 3</u>

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

1. **Variables:**
   a. Outcome: Difference of time spent on care activities (i.e. the difference in the men's share of care work vs. the women's share of care work during the week [care work on weekends was not asked in every survey year]; pli0044_h from dataset pl)
   b. Central IV: Having as first child a daughter (1/0) constructed by using biobirth data on first-born sex (kidsex01; biobirth). The dummy takes the value of 1 if first kid's sex is identified as female and 0 as male.
   c. Controls: see preamble. We estimate a regression of heterosexual couples from the women's perspective (see 2). Additionally, to the standard controls (see preamble) we also include the partner's education, age, and employment status. Please note that we do not

control for gender, as we have only women in our sample. Further, we account for labour market status in a more nuanced manner than in the other estimations, since the time spent (or not spend) at the labour market is deeply entangled with the time one can spent (or not spend) on care work.

- Employment status (full-time employment [=1] / part-time employment [=2] / not employed [= 3] / other [= 4]):
  - recode pgemplst (1 = 1 "full-time") (2 = 2 "part-time") (5 = 3 "not employed") (3 4 6 = 4 "other"), gen(work)

2. **Inclusion/exclusion criteria:**
   Observations: The analysis sample consists of females from private households (inlist(pop,1,2,5,6)) who conducted an individual questionnaire (inrange(netto,10,19)) between 2000 and 2020 whose first-born child was born 1 to 12 months before the interview and was not part of a multiple birth. Furthermore, we restrict our sample to women in heterosexual relationships in the SOEP survey, as previous research has found that division of paid and unpaid labor is more equally shared in same-sex households (see e.g. Goldberg et al. 2012). We assume missingness on covariates to be completely at random with regard to the association of outcome and gender of first-born child thus excluding observations with missing values on either of the covariates (unless specifically coded as values, see preamble), auxiliary information used for variable construction (interview month, birth year and birth month, etc.) or dependent variable (i.e., list-wise deletion). Note that we do have a few duplicate observations in the sample, i.e. respondents that were interviewed two times during the first twelve months after giving birth, in this case we chose to keep only the first observation in the sample.

3. **Exact analysis:**
   a. Method: OLS regression
   b. Standard error adjustment: none

4. **Interpretation of results:**
   We are interested in both the direction and statistical significance of the "daughter" effect. To preliminarily confirm the hypothesis, the estimated coefficient must be positive and statistically significant at the 5% level or below.

Hypothesis 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).
That is, denoting X as a measure of support for gender equality,

X for male respondents with daughters − X for male respondents without daughters
>
X for female respondents with daughters − X for female respondents without daughters

1. **Variables:**
    a. Outcome: Gender equality measured by support for the statement “It’s best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.” (var: plh0308_v2, data: pl). Answer scale is a 7-point likert scale ranging from 1 completely disagree to 7 completely agree.
    b. Central IV: Having as first child a daughter (1/0) constructed by using biobirth data on first-born sex (kidsex01; biobirth). The dummy takes the value of 1 if first kid’s sex is identified as female and 0 as male. Due to limited case numbers, we extend the sample to include all parents whose first child was born between 1 and 24 months prior to the 2018 interview—rather than restricting the window to 1–12 months, as done in the other hypothesis tests. We account for this longer time frame by including a control for months passed since first-born birth (msb). However, this is not without limitations: if the treatment effect evolves over time, part of it may be captured by the msb variable, potentially attenuating the estimated effect of having a daughter.
    c. Controls: see preamble. Please note that we do not control for gender, as we ran gender separated models.

1. **Inclusion/exclusion criteria:**
    Observations: The analysis sample consists of individuals from private households (inlist(pop,1,2,5,6)) who conducted an individual questionnaire (inrange(netto,10,19)) in 2018 (syear==2018) whose first-born child was born up to 24 months before the interview and was not part of a multiple birth. We assume missingness on covariates to be completely at random with regard to the association of outcome and gender of first-born child thus excluding observations with missing values on either of the covariates (unless specifically coded as values, see preamble), auxiliary information used for variable construction (interview month, birth year and birth month, etc.) or dependent variable (i.e., list-wise deletion).

2. **Exact analysis:**
    a. Method: We calculate two OLS regressions separately for male respondents and female respondents. Based on these two regressions, we estimate the difference in the daughter effect for male and female respondents simultaneously (using the suest command) and calculate a χ^2 Wald test for equality of daughter coefficients of male and female respondents to test for statistical significance of the difference.

b. Standard error adjustment: clustered robust standard errors at the household level (hid).

5. **Interpretation of results:**
   In the OLS regression stratified by gender, each "daughter" coefficient captures the difference in gender equality attitudes between parents of daughters and parents of sons—separately for women and men. The hypothesis under investigation posits that this difference is larger for men than for women. We are therefore interested in both the direction and statistical significance of the difference between the male and female coefficients. A positive difference would indicate a stronger daughter effect among male respondents. To assess whether this difference is statistically meaningful, we perform a χ^2 Wald test. The hypothesis is considered preliminarily confirmed if we find a statistically significant effect at the 5% level or below.

References

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